

Menstrual Products and Women's Health: Research Review

Compiled from peer-reviewed studies, government sources, and health organizations.
Mapped to the seven products tested in this project.

Products Covered

Product	Type	Category
Naturella Pad	Commercial disposable pad	Commercial
Always Platinum	Commercial disposable pad	Commercial
Ria Ultra Pad	Commercial disposable pad	Commercial
Ria Tampon	Conventional tampon	Tampon
o.b. Tampon	Conventional tampon (no applicator)	Tampon
Jessa Cotton Pad	Organic cotton pad	Organic
Jessa Cloth Pad	Reusable cloth pad	Cloth

1. Bacterial Growth Risk

Mechanism

Menstrual blood provides a nutrient-rich medium for bacterial proliferation. The risk depends on whether the product is worn externally (pads) or internally (tampons), how long it is worn, and the material composition.

By Product Type

Tampons (Ria Tampon, o.b. Tampon) — HIGHEST RISK - Tampons introduce oxygen into the vagina, creating an aerobic environment favorable for *Staphylococcus aureus* growth (Clue, 2024).
- The 1979–1980 TSS outbreak in the US recorded 1,365 cases linked primarily to the Rely tampon, which used synthetic super-absorbent materials. Risk was 6–12 times higher than current levels (Clue, 2024; CDC).
- Menstruation raises vaginal pH (normally ~3.8–4.5), making it less acidic and more hospitable to bacterial multiplication (Clue, 2024).
- Healthcare providers recommend replacing tampons every 4–8 hours to minimize bacterial growth (Clue, 2024; Healthline, 2022).

Commercial Disposable Pads (Naturella, Always Platinum, Ria Ultra Pad) — MODERATE RISK - Worn externally, so they do not directly alter the vaginal environment or introduce oxygen.
- However, prolonged wear without changing creates a warm, moist environment against the vulva that can promote bacterial and fungal growth on the skin surface.
- Synthetic materials (polyethylene, polypropylene topsheets) in commercial pads may trap moisture more than natural fibers, potentially increasing surface bacterial growth.

Organic Cotton Pad (Jessa Cotton Pad) — LOWER RISK - Organic cotton is more breathable than synthetic-blend commercial pads, reducing moisture retention.
- No synthetic superabsorbent polymers (SAP) that could harbor bacteria.
- Still external-use, so no direct vaginal bacterial introduction.

Reusable Cloth Pad (Jessa Cloth Pad) — VARIABLE RISK - Cloth is breathable and natural, but improper washing between uses can lead to bacterial colonization in the fabric. - If not sterilized adequately (e.g., hot water wash, sun-drying), residual bacteria from previous cycles can multiply. - With proper hygiene, risk is comparable to or lower than disposable pads.

Key Finding

Internal products (tampons) carry inherently higher bacterial growth risk due to direct vaginal contact and oxygen introduction. External pads are lower risk, but material breathability and change frequency matter significantly.

2. Toxic Shock Syndrome (TSS)

What Is TSS?

TSS is a rare but life-threatening condition caused by toxins released by *Staphylococcus aureus* or *Streptococcus pyogenes* bacteria. Current incidence is approximately **1 in 100,000** menstruating people per year in the United States (Clue, 2024).

By Product Type

Tampons (Ria Tampon, o.b. Tampon) — PRIMARY ASSOCIATION - Tampons are the most common factor in menstrually-associated TSS (Clue, 2024). - Higher-absorbency tampons pose greater risk because they can cause micro-tears in the vaginal wall during insertion/removal, providing entry points for toxins into the bloodstream (CDC; Healthline, 2022). - The FDA now requires standardized absorbency labeling (light, regular, super, ultra) so users can choose the minimum absorbency needed (Colorado University, 2026). - Both Ria and o.b. tampons are conventional (cotton/rayon blend), which carry the standard TSS risk profile.

All Pad Types (Naturella, Always Platinum, Ria Ultra, Jessa Cotton, Jessa Cloth) — NO TSS RISK - TSS from pads has not been documented because pads are external and do not introduce bacteria into the vagina or create the anaerobic-to-aerobic shift that promotes toxin production. - This is a critical differentiator: pads eliminate TSS risk entirely.

Symptoms of TSS

Fever, dangerously low blood pressure, rash and skin peeling, organ dysfunction, diarrhea, muscle pain, confusion. Requires immediate medical attention (Clue, 2024).

Key Finding

TSS risk is exclusive to internal products (tampons, and very rarely menstrual cups). All five pad products in this project carry zero TSS risk.

3. Skin Irritation

By Product Type

Commercial Pads (Naturella, Always Platinum, Ria Ultra Pad) — MODERATE RISK - Synthetic topsheets, adhesives, and fragrances are common irritants. - Scented pads in particular can irritate sensitive vulvar skin and cause contact dermatitis (Brigham and Women's Hospital). - The plastic backing prevents airflow, creating a humid microenvironment that can cause chafing and irritation during prolonged wear.

Organic Cotton Pad (Jessa Cotton Pad) — LOWEST RISK - Organic cotton is free from synthetic fragrances, dyes, and chemical bleaching agents. - Natural fiber is softer and more breathable, reducing friction and moisture buildup. - Best option for users with sensitive skin or contact dermatitis history.

Reusable Cloth Pad (Jessa Cloth Pad) — LOW RISK (with caveats) - Natural cotton cloth is gentle on skin and highly breathable. - Risk arises from detergent residues if not rinsed thoroughly — some laundry detergents contain fragrances and chemicals that can irritate. - With fragrance-free detergent and thorough rinsing, irritation risk is minimal.

Tampons (Ria Tampon, o.b. Tampon) — LOW TO MODERATE RISK - Internal use means less direct skin contact with the vulva. - However, the applicator (Ria) or insertion process (o.b., no applicator) can cause friction-related irritation. - Rayon fibers in conventional tampons may cause more irritation than 100% cotton (Healthline, 2022). - The o.b. tampon's smooth covering (no applicator) may reduce external irritation but requires finger insertion, which can introduce bacteria if hands are not clean.

Key Finding

Organic and cloth pads present the lowest skin irritation risk. Commercial pads with synthetic materials and fragrances present the highest. Tampons present moderate risk primarily from insertion friction rather than surface irritation.

4. Chemical Exposure

Heavy Metals in Tampons

- A 2024 study (Shearston et al., *Environment International*) detected **16 heavy metals** across 14 tampon brands, including **lead, arsenic, cadmium, and mercury** (Brookings, 2024; Colorado University, 2026).
- **Lead was found in all 30 tampons tested** in the study (Colorado University, 2026).
- Heavy metals are absorbed by cotton plants from soil (natural or polluted) and carry through to the final product.
- The FDA stated there are “no safety concerns associated with tampon use and contaminant exposure” based on a literature review, but acknowledged the need for further research (BAHP, 2024).

VOCs and Phthalates in Pads

- A 2019 study found sanitary pads contain **volatile organic compounds (VOCs) and phthalates**, both associated with menstrual irregularities and cancer risk (Brookings, 2024).
- VOCs may be added as fragrances, adhesives, or other product components and can be absorbed through the vaginal/vulvar mucosa (Colorado University, 2026).
- Phthalates leach from plastic packaging or are added as fragrance carriers (Colorado University, 2026).

PFAS (“Forever Chemicals”) in Reusable Products

- A 2025 study (Notre Dame/University of Notre Dame, published in *Environmental Science and Technology Letters*) found PFAS in reusable menstrual products:
 - **33% of period underwear** had intentionally added PFAS
 - **25% of reusable pads** had intentionally added PFAS
 - PFAS levels reached **parts per million** (not just parts per trillion) — described as “very, very high” (CNN, 2025; ACS, 2025).
- Neutral PFAS (newly measurable) are more easily absorbed through skin and mucous membranes because they move through fatty tissues more readily (CNN, 2025).
- PFAS are known endocrine disruptors linked to cancer, obesity, high cholesterol, decreased fertility, low birth weight, and hormone disruption (EPA via CNN, 2025).
- Importantly, only a subset of reusable products had high PFAS levels, meaning PFAS-free manufacturing is feasible (CNN, 2025).

Dioxins from Bleaching

- Chlorine bleaching of cotton produces dioxins, associated with cancer and endocrine disruption (Colorado University, 2026).
- The FDA recommends (but does not require) that tampons and pads should not contain specific dioxin byproducts (Colorado University, 2026).
- Non-chlorine bleaching methods reduce dioxin formation.

By Product

Product	Chemical Exposure Risk	Key Concerns
Naturella Pad	Moderate	VOCs, phthalates from synthetic materials, possible fragrance chemicals
Always Platinum	Moderate	VOCs, phthalates, synthetic polymers, adhesives
Ria Ultra Pad	Moderate	VOCs, phthalates, synthetic materials
Ria Tampon	Higher	Heavy metals (Pb, As, Cd), dioxins from bleaching, rayon fibers
o.b. Tampon	Higher	Heavy metals (Pb, As, Cd), dioxins from bleaching

Product	Chemical Exposure Risk	Key Concerns
Jessa Cotton Pad	Lower	Organic cotton reduces pesticide residue; still may contain trace environmental metals
Jessa Cloth Pad	Variable	PFAS risk if waterproofing layer uses treated fabric; detergent chemical residue from washing

Regulatory Context

- Tampons, menstrual cups, and scented pads are FDA Class II medical devices (moderate risk); unscented pads are Class I (low risk) (Colorado University, 2026).
- The FDA does **not mandate regular testing** for toxic chemicals in menstrual products (Brookings, 2024).
- Vermont (2024), California (2024–2025), and New York (2025) have passed or are passing laws banning PFAS and other toxic chemicals from menstrual products (Colorado University, 2026).

Key Finding

All commercial disposable products carry some chemical exposure risk. Organic cotton products reduce but do not eliminate risk (environmental heavy metals are unavoidable). Reusable cloth products carry a PFAS risk if the waterproof layer is chemically treated, but this is manufacturer-dependent and not inherent to the product type.

Summary Comparison Table

Health Dimension	Naturella	Always Platinum	Ria Ultra Pad	Ria Tampon	o.b. Tampon	Jessa Cotton	Jessa Cloth
Bacterial Growth	Moderate	Moderate	Moderate	High	High	Low	Variable*
TSS Risk	None	None	None	Present	Present	None	None
Skin Irritation	Moderate	Moderate	Moderate	Low–Mod	Low–Mod	Low	Low*
Chemical Exposure	Moderate	Moderate	Moderate	High	High	Low	Variable**

* Depends on washing hygiene ** Depends on PFAS treatment of waterproof layer and detergent used

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